

Capstone Scope Document — KijaniKiosk

Track: Track A — Infrastructure-First

Problem Statement

KijaniKiosk currently lacks a controlled multi-environment delivery path that validates application changes in an isolated staging environment before production promotion. This creates a risk that faulty changes can reach production without first being validated against staging configuration, application health checks, and runtime behaviour. The capstone will introduce a reproducible staging environment, automated staging validation, monitoring, and an approval-controlled production deployment while integrating the Week 10 serverless receipt chain.

What I Will Build

- Staging infrastructure:** Terraform provisions an isolated `kijani-staging` namespace and Ansible configures it.
- Staging application:** `kk-payments` uses the existing Deployment design with staging-specific configuration and a different `DB_HOST` from production.
- Controlled CI/CD:** Jenkins deploys to staging on merge to `main`, runs a smoke test, then presents the production approval gate.
- Runtime monitoring:** A committed monitoring signal measures `kk-payments` error rate and is demonstrated during testing.
- Serverless integration:** Staging payments write to `kk-payments-receipts-staging` and the Week 10 receipt chain processes the event.

What Is Out of Scope

- **Track B** `kk-analytics`: excluded because this project follows Track A.
- **Rewriting** `kk-payments`: excluded because the capstone extends the existing application rather than replacing its business logic.
- **Full cloud migration**: excluded because the objective is controlled staging-to-production delivery and integration of the existing platform.

Success Criteria

- A merge to `main` triggers Jenkins, deploys to `kijani-staging`, and passes a smoke test before production approval appears.
- An authorized human can approve production after successful staging validation, with approval reason and approver recorded in pipeline evidence.
- A staging payment creates a receipt event in `kk-payments-receipts-staging` and the Week 10 serverless receipt chain processes it successfully.

Architecture

KijaniKiosk Capstone — Track A Architecture

Automated staging validation → human approval → production

